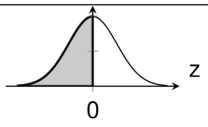
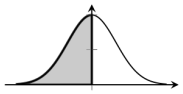
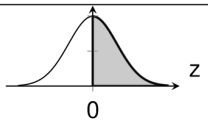
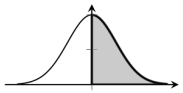
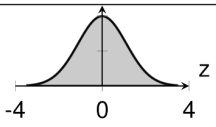
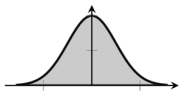
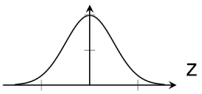
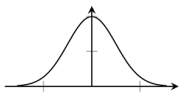
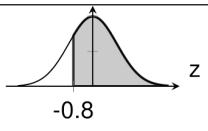
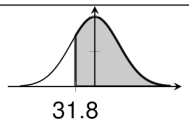
The Normal Distribution

LLE – Mathematics and Statistics

1. These questions concern the standard normal distribution, Z, which has a mean $\mu = 0$ and a variance $\sigma^2 = 1$, so you can write $Z \sim N(0, 1)$. You will also consider the distribution of the length of donkey's tails, $D \sim N(35, 16)$, that is the normal distribution with mean $\mu = 35$ cm and variance $\sigma^2 = 16$ cm^2. Fill in the blanks using the information throughout the table below. No tables or calculators are required.

$P(Z < 0) = 0.5$		$P(D < 35\text{cm}) =$	
$P(Z > 0) =$			
$P(Z > -4) \approx$			
$P(Z >$ $) \approx 0$		$P(D >$ $) \approx$	
$P(Z < 0.8) = 0.788$		$P(D < 38.2) =$	
$P(Z > 0.8) =$		$P(D > 38.2) =$	
$P(Z <$ $) = 0.212$		$P(D <$ $) = 0.212$	
$P(-0.8 < Z < 0.8) =$		$P(31.8 < D < 38.2) =$	
			

2. Calculate the standardised score z and the non-standardised score x for each of the distributions below

Distribution for X	**x-value (raw value)**	**z-value (standardised value)**
$X \sim N(30, 25)$	35	
$X \sim N(20, 25)$	20	
$X \sim N(100, 256)$		1.5
$X \sim N(100, 256)$	140	
$X \sim N(50, 6^2)$	29	

3. Given the information about the population distribution, find the probabilities requested by first standardising the raw scores. Make sketches of the X and Z distributions with the probability areas shaded.

   (a) $X \sim N(80, 25)$

      i. $P(X < 85) = P(Z < ?) = ?$ (standardise the value of 85)

      ii. $P(X > 70)$

      iii. $P(78 < X < 82)$

   (b) $X \sim N(150, 156.25)$

      i. $P(X < 143.75)$

      ii. $P(X > 180)$

      iii. $P(145 < X < 170)$

4. Given the information about the population distribution, find the values that are represented by the following probabilities/proportions. Give your answers correct to 1 decimal place. You will need to find the z-score from the probability given and then transform this back to a raw score. Make sketches to show the situations.

   (a) $X \sim N(100, 16)$

      i. $P(X < r) = 0.9772$

      ii. $P(X > s) = 0.9772$

      iii. $P(X < t) = 0.35$

(b) $X \sim N(20, 4)$

i. $P(X < r) = 0.9772$

ii. $P(X > t) = 65\%$

5. A hot drinks machine is known to dispense liquid in a volume that is normally distributed with a mean of 250 ml and a standard deviation of 5 ml, $V \sim N(250, 25)$

   (a) What proportion of drinks are expected to be dispensed that are over 254 ml?

   (b) What proportion of drinks that are expected to be under 246 ml in volume.

   (c) Use your answers and sketches to parts b and c to work out the probability that a drink is dispensed that is between 246 ml and 254 ml.

   (d) What volume are 75% of drinks less than?

   (e) You find that your cup overflows 2% of the time. How big is your cup?